

Why Employees Leave Salifort Motors — and How to Keep Them

Analysis of HR survey data for 11,991 unique employees and a predictive model that flags attrition risk before it happens.

16.6%

employees left

92%

of leavers caught by the model

0.97

model AUC (hold-out)

~180

current employees at risk

THE PROBLEM

High turnover is costly

1,991 of 11,991 employees (16.6%) left the company. Every departure means new hiring, onboarding, and training costs — plus lost experience. Leadership needs to understand why people leave and see who is at risk early.

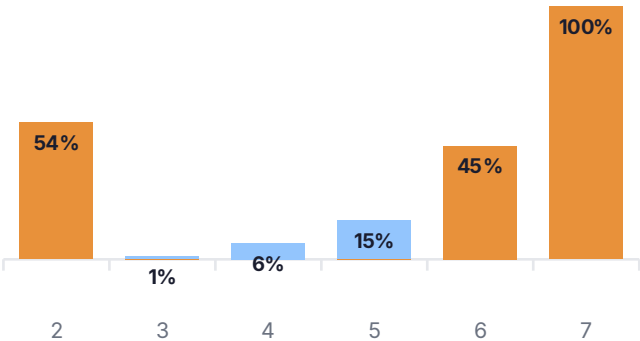
OUR APPROACH

Seven models, one practical winner

Cleaned the data (removed 3,008 duplicates), ran EDA and hypothesis tests, and compared logistic regression, decision tree, random forest, and XGBoost with 5-fold cross-validation. **Selected: random forest without survey data** — it can run monthly on HRIS data.

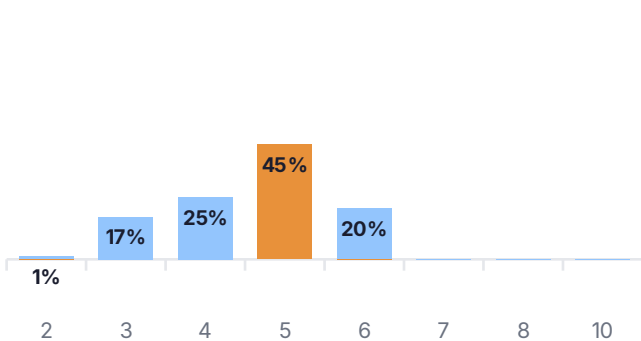
Attrition rate by number of projects

Sweet spot: 3–4 projects. At 7 projects, everyone left.



Attrition rate by tenure (years)

Risk peaks in years 4–5; nobody leaves after year 6.



Key insight: three profiles explain 93% of departures — burnout (6–7 projects, 250–310 h/month), underloaded (2 projects, ~145 h), and overworked high performers (evaluation ≈0.9, 4–5 years). The median employee works ~200 h/month vs. a ~167 h standard, and only 1.7% were promoted in five years. Department is not a driver (12–19% everywhere).

RECOMMENDATIONS

Cap the workload

Max 4–5 projects per person

Flag >200 h/month to managers; pay for overtime or give comp time.

Invest in years 3–5

Promotion rate today: 1.7%

Run a pay and career review for high performers with 4+ years; reward results, not hours.

Score risk monthly

Pilot in 1–2 departments

Use the model to start supportive check-ins — never for penalties. Measure impact vs. a control group.